



*I consider myself as a Versatile I&C Engineer with a history of rapid career progression. Focused on delivering high-impact solutions for the evolving energy industry.*

**General Context**

**PROFESSIONAL EXPERIENCE WORLD WIDE**

*Transitioning into the European engineering market with a solid background in large-scale EPC projects in Mexico and Spain.*

*Experienced in international power generation projects following EU standards, with a progressive career shift from instrumentation engineering toward a specialization in control systems, operating logic, and safety systems for combined cycle power plants and LNG terminals.*

Enterprise

Enterprise Location

Role Description

Period of time

Project

Project Location

Key Responsibilities / Activities

Project

Project Location

Key Responsibilities / Activities

Project

Project Location

Key Responsibilities / Activities

Project

Project Location

Key Responsibilities / Activities

**TSK**

**Gijón, Asturias**

**Specialist Control Engineer:**

Focused on combined cycle power plants, LNG terminals, and associated energy facilities. Key responsibilities include the development of control logic and safety systems, the preparation of functional documentation, and coordinating client acceptance testing (FAT).

December 2023 – Now

**2 Combined Cycles (500 MW and 450 MW)**

Dominican Republic

- Technical review of vendor documentation.
- Validation of alarm lists, functional descriptions, and I/O signal lists.
- Support for the design and review of DCS HMI/operating screens.

Development of red-lines and as-built documentation.

**Combined Cycle (450 MW)**

Mexico

- Commissioning
- Technical review of vendor documentation.
- Validation of alarm lists, functional descriptions, and I/O signal lists.
- Support for the design and review of DCS HMI/operating screens.
- Development of red-lines and as-built documentation.

**Mugardos LNG Terminal Expansion — Reganosa**

Galicia, Spain

- Development of cause-and-effect matrix.
- Preparation of functional descriptions and control logic.
- Definition of interlocks and emergency shutdown (ESD) trips.
- Actualization of alarm lists and setpoints.
- Adequation of emergency shutdown (ESD) philosophies.
- Technical review of vendor documentation.
- Issuance and control of engineering documentation.
- Support for Factory Acceptance Tests (FAT).
- Operated an industrial BOG reciprocating compressor.
- Developed and maintained the daily site log, accurately tracking project milestones, operational data, and technical incidents.
- Conducted performance testing on-site to validate systems integrity and operational automatic sequences.
- Development of red-lines and as-built documentation.

**Comined Cycle (~150 MW)**

Dominican Republic

- Control engineering development based on KKS identification standards.
- Preparation of system functional descriptions.
- Development of alarm summaries and setpoint lists.
- Definition of control logic, interlocks, and safety functions.
- Technical review of vendor documentation.
- Support for DCS HMI/operating display design.



- Participation in Factory Acceptance Tests (FAT) with client presence.

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enterprise                        | <b>Intecsa Industrial</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Enterprise Location               | <b>Madrid, Spain</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Role description                  | <b>Instrumentation and Control Engineer:</b><br>Participation in the detailed engineering phase of a combined cycle power generation project, working under KKS identification standards and regulations within a European EPC environment, adapting to distinct workflows and methodologies compared to Oil & Gas projects                                                                                                                                                                                                                                                                                                          |
| Period of time                    | September 2023 – November 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Project                           | <b>Combined Cycle (~450 MW)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Project Location                  | Kazajistan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Key Responsibilities / Activities | <ul style="list-style-type: none"><li>• Preparation and review of instrumentation datasheets in SmartPlant Instrumentation (SPI).</li><li>• Sizing of pressure safety valves (PSVs) in compliance with contractual and piping specifications.</li><li>• Technical analysis of material compatibility based on process conditions.</li><li>• Issuance of Requests for Quotation (RFQ) and technical bid evaluation for PSV procurement.</li><li>• Review of P&amp;IDs (Piping and Instrumentation Diagrams) under KKS standards.</li><li>• Adaptation to European standards for documentation, nomenclature, and workflows.</li></ul> |
| Professional Evolution            | Consolidated expertise in control engineering for the power sector, specializing in KKS identification, functional logic, and safety systems. Proven track record in vendor management and customer acceptance testing, demonstrating a high degree of technical independence, consistency, and professional maturity in large-scale international EPC projects.                                                                                                                                                                                                                                                                     |



Enterprise  
Enterprise Location  
Role Description  
  
  
  
Period of time  
Project  
Project Location  
Key Responsibilities / Activities

**ICA Fluor Daniel**  
**Mexico City, Mexico**  
**Sr Control Engineer:**

Involved in a large-scale EPC project for the detailed engineering and subsequent construction of a Combined Distillation Unit (CDU/VDU) with a capacity exceeding 300,000 barrels per day across two trains, along with associated projects within the same industrial complex. My role progressed from detailed engineering activities to key responsibilities in materials management, construction support, and on-site commissioning testing  
2019 – 2023

**Combined Distillation Unit (CDU)**  
Mexico

- **Instrumentation Engineering – Detailed Design Phase**
  - Review and enhancement of P&IDs (Piping & Instrumentation Diagrams) from basic engineering.
  - Integration of local instrumentation, control systems, control strategies, and redundancy schemes.
  - Technical support for specialists in defining instrumentation bulk volumes.
  - Development of the Master Instrument Index, structured by WBS (Work Breakdown Structure) and CWP (Construction Work Package).
  - Consolidation of instrumentation lists exceeding 30,000 tags for the entire facility.
- **Procurement Support – Critical & Bulk Equipment**
  - Preparation of Technical Bid Tabulations (TBT) for high-volume instrumentation, including:
    - Pressure gauges (1,500+ units).
    - Pressure transmitters (approx. 500 units).
    - Sampling systems and tapping points (35 units).
  - Technical evaluation of compliance with project specifications and client requirements.
- **Engineering Corporate Tools**
  - Comprehensive management of instrumentation data in SmartPlant Instrumentation (SPI – Hexagon).
  - Loading and maintenance of instrument tags, signal types, and datasheets within the database.
  - Optimization of datasheet generation through:
    - Advanced use of the Browser module.
    - Configuration of views and EDE (Enhanced Data Entity) modules for bulk data loading and export.
  - Substantial improvement in documentation turnaround times.
- **Engineering Standardization & Improvement**
  - Development and standardization of installation details (hook-ups) per instrument type.
  - Definition of distinct hook-ups based on:
    - Instrument type
    - NACE and non-NACE compliant materials.
    - Special alloys (Monel, Inconel, Hastelloy C).
    - Special accessories (Tubing, chemical seals, capillary tubes)
  - Identification and development of details for Electric Heat Tracing (EHT) and steam tracing systems.
  - Implementation of documentation methodologies subsequently replicated across other projects.

Associated Project  
Key Responsibilities / Activities

**Storage Tank Area – CDU/VDU**

- Support for I/O signal lists and associated instrumentation.
- Development of schematics and installation hook-ups.
- Sulphur Recovery Unit (SRU)
- Instrumentation engineering support for small-scale facility projects.
- Proficiency in Material Manager software for procurement and inventory control.
- Complete data entry for project material take-offs (MTO) into the system.



Associated Project  
Key Responsibilities / Activities

### **Materials Management & Multi-Plant Support**

- Management and data entry of instrumentation materials for seven plants within the complex.
- Consolidation, correction, and updating of material volumes.
- Issuing periodic material status reports and logistic scenario analyses.
- Technical coordination for material redistribution and inter-plant transfers based on site requirements.

Associated Project  
Key Responsibilities / Activities

### **Construction and Commissioning support (more than 2 years)**

- On-site support during the construction and pre-commissioning phases.
- Preparation and updating of as-built documentation and red-lines.
- Comprehensive instrumentation materials management, including:
  - Purchase Requisitions (PR).
  - Field distribution and inventory control.
  - Material transfers between different work fronts.
- Preparation of weekly progress reports covering quantities, costs, and lead times.
- Support for on-site loop checks.
- Participation in functional testing for oxygen and analyzers in crude and atmospheric residue furnace stacks.
- Supervision of commissioning activities and work at height.
- Support during final startup stages, up to the initial firing of the main furnaces.

Professional Progression

Earned two promotions within a single year as a result of consistently increasing technical responsibilities and high-level engagement with project management. This role involved critical decision-making regarding material volumes, cost control, logistical strategies, and project execution priorities.



Enterprise  
Enterprise Location  
Role Description  
  
  
Period of time  
Project  
Project Location  
Key Responsibilities / Activities

## **TURBO-MEX (TBX)**

**Mexico City, Mexico**

### **Jr Control Engineer:**

Contributed to Instrumentation and Control (I&C) engineering projects within a mid-sized EPC firm, working in lean, multi-disciplinary teams. This role provided high exposure to the entire project lifecycle, from technical specification reviews and instrumentation procurement to workshop activities, pre-commissioning, and functional testing

2018 – 2019

### **9 Test Separator Skids for Unmanned Offshore Platforms**

Mexico

- **Technical Management & Procurement Support**
  - Analysis of client technical specifications for the correct selection of instrumentation and control equipment.
  - Request for Quotation (RFQ) and technical bid evaluation of instrumentation vendors.
  - Verification of compliance with applicable standards and certifications, including NACE MR0175 / ISO 15156 and NEMA 4X / NEMA 7 ratings.
  - Preparation of Technical Bid Tabulations (TBT) to support the procurement process.
  - Technical support to the procurement department during vendor and equipment selection.
- **Instrumentation Engineering (Detailed Design)**
  - Development and maintenance of the Project Instrument Index.
  - Preparation of instrument datasheets based on process and operating conditions.
  - Definition of measuring ranges, materials, and technical specifications.
  - Development of installation hook-ups (mechanical and electrical) drawings.
  - Preparation of technical documentation for review and Approved for Construction (AFC) release.
- **Workshop, Assembly & Testing Activities**
  - Supervision and support during junction box (JB) assembly.
  - Verification of instrumentation wiring and electrical circuits.
  - Execution of continuity tests and signal validation.
  - Technical support during skid assembly and coordination with workshop personnel.
- **Functional Testing & Pre-commissioning**
  - Supervision of functional testing for instrumentation and associated systems.
  - Validation of control valves, alarms, and measurement instruments through simulated values.
  - Technical support during flow computers configuration and testing.
  - Support in the review of functional descriptions and validation of operational behavior.
- **Standalone Photovoltaic (PV) Systems**
  - Design and sizing of PV systems for autonomous skid power supply.
  - Electrical load calculations under various operating conditions.
  - Sizing of battery banks, considering capacity, charging rates, and days of autonomy.
  - Technical selection of batteries and photovoltaic panels.
  - Execution and supervision of Factory Acceptance Tests (FAT) for batteries and PV cells.
  - Testing of recharge times, autonomy, and system consumption across different operational scenarios.



Project  
Project Location  
Key Responsibilities / Activities

## 8 Sour Gas Scrubbing Skids

Mexico

- Development of local instrumentation engineering for various skid sizes and configurations.
- Development of instrument lists and indexes.
- Preparation of instrument datasheets specifically for corrosive media environments.
- Preparation of Technical Bid Tabulations (TBT) for vendor proposal evaluation.
- Development of mechanical and electrical installation hook-ups (typical drawings).
- Supervision and technical support during workshop instrumentation assembly.
- Technical coordination with other disciplines throughout the assembly process.

Professional Progression

Started my career in a fast-paced EPC environment, gaining early exposure to the full lifecycle of modular Skid units (Test Separators and Sour Gas Scrubbers). This foundational role allowed me to bridge the gap between detailed engineering and hands-on workshop assembly, developing a strong technical grasp of instrumentation selection for corrosive environments and autonomous power systems